3.0 GENERAL LOADING INFORMATION

3.1 Overview

3.1.1 This section reprints mandatory rules contained in AAR Circular No. 42-N. For the most up-to-date rules, see Circular No. 42-N or supplements thereto.

3.1.2 Most paper shipped via rail is wrapped. Generally, liner board, corrugating medium, and similar types of paper are not wrapped, and the outer plies are considered the protective wrap for these types of rolls. Newsprint and printing papers are in most cases wrapped to protect all of the paper on the rolls.

3.1.3 Load, block, or brace commodities tightly lengthwise and crosswise to eliminate all void spaces, which are the primary reasons for damage. Take up any void spaces remaining in a car. Use blocking, fillers, and other suitable materials, and secure them in accordance with the methods outlined in this guide and other guides.

3.1.4 Load and secure lading to permit unloading from either side of the railcar, except when dimensions of individual units of freight prohibit unloading from either side of the car. Loads that cannot be unloaded from either side of the railcar may incur additional switching charges.

3.1.5 When loading, segregate and protect commodities that may cross contaminate.

3.1.6 The ends of boxcars are designed to withstand a horizontal force induced by the lading without exceeding the yield strength of the material (see Table 3.1).

Table 3.1 Boxcar endwall strength

Nominal Capacity of Car (ton)	Total Force on Endwall (lb)	Percentage of Total Force Uniformly Distributed	
		Top Half	Bottom Half
50	100,000	35–45	65–55
70	220,000	35–45	65–55
100 (263,000 lb)	260,000	35–45	65–55
100 (286,000 lb)	284,000	35–45	65–55

3.2 Maximum Load Weight

3.2.1 The weight of the load in the car must not exceed the load limit stenciled on the car or as listed in UMLER. If there is a discrepancy between the two load limits, use the lower of the two load limits and report the discrepancy to the serving railroad.

3.2.2 The weight of the load on one truck must not exceed one-half of the load limit stenciled on the car or as listed in UMLER. If there is a discrepancy between the two load limits, use the lower of the two load limits and report the discrepancy to the serving railroad.

3.3 Distribution of Weight Crosswise in Cars

3.3.1 The load must be located so that the weight along both sides of the car is approximately equal for the entire length of the load.

3.3.2 When the load is such that it cannot be placed to obtain equal distribution of weight crosswise of the car, use properly secured and suitable ballast to equalize the weight.

3.3.3 In boxcars, lading must be secured to prevent tipping or moving toward the sides of the car where the vacant space across the car exceeds the following:

- An aggregate of 18 in. crosswise of car
- Vacant crosswise space of less than 18 in. as may be specified in guides covering methods for loading, bracing, and blocking carload shipments of individual commodities

3.4 Distribution of Weight Lengthwise in Cars

Evenly distribute the weight of the load lengthwise in the railcar. Fill all lengthwise space with lading, filler material, or appropriate blocking and bracing materials.

3.5 Center of Gravity

3.5.1 The combined center of gravity of a railcar and its contents must not exceed 98 in. above top of rail. In closed cars, there is no practical possibility of exceeding this center of gravity limitation, except in cars that exceed Plate C dimensions. See paragraph 11.0, "Equipment Diagrams for Unrestricted Interchange Service."

3.5.2 Cars exceeding Plate C dimensions may extend to 17 ft above top of rail. Certain lading, such as rolled paper loaded two layers high, may result in excessive combined center of gravity dimensions. Shippers must calculate the combined center of gravity of the railcar and contents whenever any part of the load will exceed 11 ft 8 in. (140 in.) in height above the car floor. Shipper's tender of billing information for such cars to the origin carrier will signify compliance with this rule. Any questions about loading limitations in cars exceeding Plate C dimensions should be addressed to the mechanical department of the origin carrier. (See paragraph 11.2, "Plate C," on page 11–2.)

3.5.3 Use the following formula to calculate the combined center of gravity:

A = Height of car floor above top of rail in inches

B = Empty center of gravity of railcar above top of rail in inches, obtainable from car owner (empty center of gravity may be stenciled on the railcar)

C = Center of gravity of load above car floor in inches

D = Height of center of gravity of load above top of rail

= A + C

E = Lightweight of railcar in pounds

F = Weight of load in pounds

NOTE: The following table may be used as a guideline when determining A in the above formula:

Weight of Loads (lb)	Spring Deflection (in.)
122,000–138,000	1.00
138,000–165,000	1.25
165,000–192,000	1.50
192,000–207,000	1.75

3.5.4 Combined center of gravity (CG) $= \frac{(B \times E) + (D \times F)}{(E + F)}$

3.5.5 EXAMPLE: Roll Paper

Load: Seventy-seven 45 in. diameter × 50 in. wide rolls each weighing 2,500 lb, loaded in 29 floor spots with two complete layers plus 19 rolls in an incomplete third layer.

Use 6 in. high risers to block the incomplete layer. Incomplete layer unitizing not shown.

NOTE: When loads consist of multiple sections or units having different unit heights and weights, each section or unit must be taken separately when calculating the CG of the load.

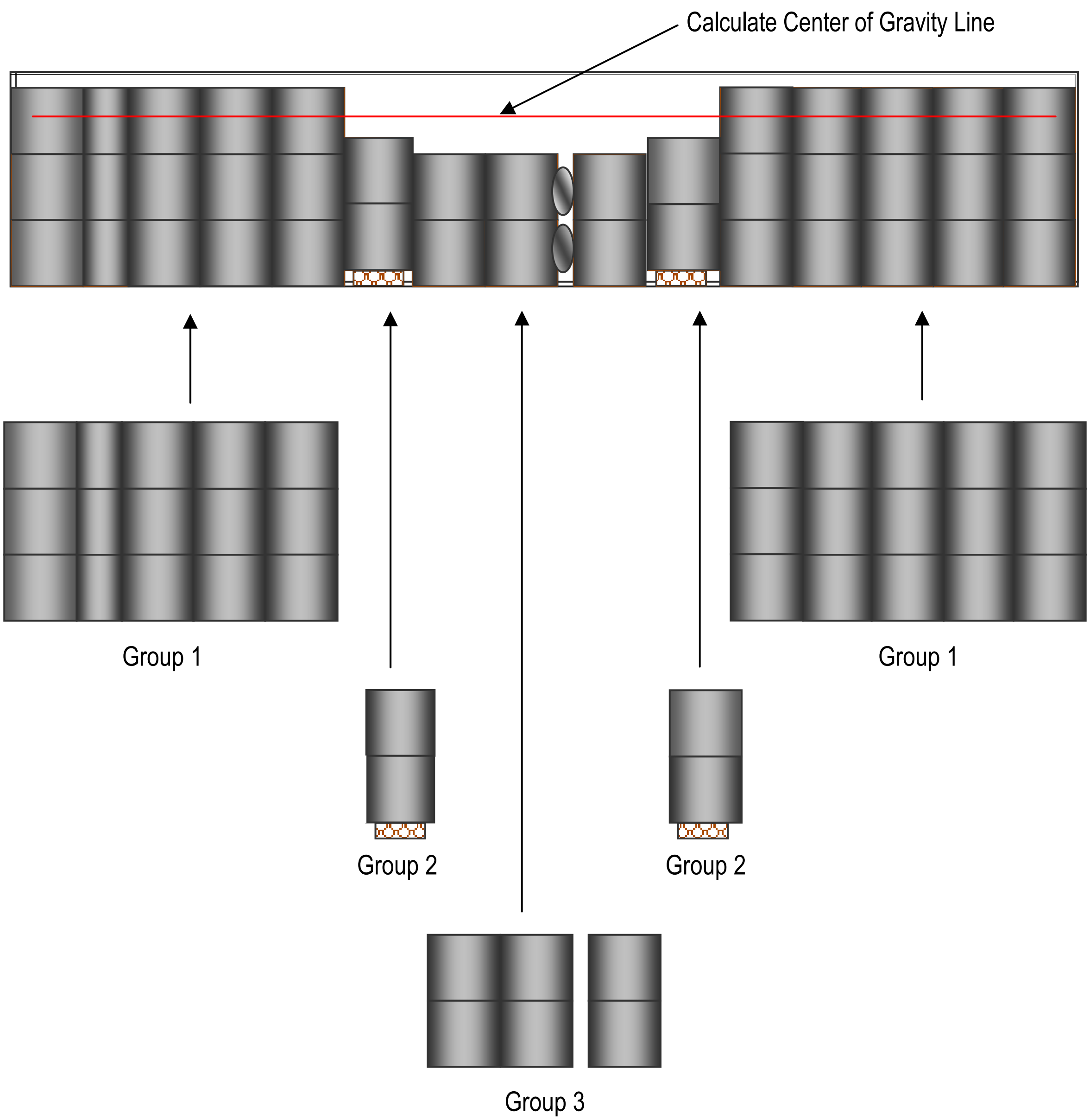

Figure 3.1 Side view of example load

All rolls in Group 1 would be considered a section of this multi-section load. All rolls in Group 2 would be considered a separate section, as would all rolls in Group 3. Riser height should also be added in when calculating the center of gravity for that section.

EXAMPLE: Roll Paper (continued)

A = 42.25 (44 in. – 1.75 spring deflection)

B = 58 in.

Group 1 = C1 = (50 in. × 3) / 2 = 75 in.
Group 2 = C2 = (50 in. × 2 + 6) / 2 = 53 in.
Group 3 = C3 = (50 in. × 2) / 2 = 50 in.

Group 1 = D1 = 75 in. + 42.25 in. = 117.25 in.
Group 2 = D2 = 53 in. + 42.25 in. = 95.25 in.
Group 3 = D3 = 50 in. + 42. 25 in. = 92.25 in.

E = 72,800 lb

Group 1 = F1 = 19 rolls × 3 layers × 2,500 lb = 142,500 lb
Group 2 = F2 = 4 rolls × 2 layers × 2,500 lb = 20,000 lb
Group 3 = F3 = 6 rolls × 2 layers × 2,500 lb = 30,000 lb

$$\text{Combined CG} = \frac{(B \times E) + (D1 \times F1) + (D2 \times F2) + (D3 \times F3)}{E + F1 + F2 + F3}$$

$$= \frac{(58 \times 72,800) + (117.25 \times 142,500) + (95.25 \times 20,000) + (95.25 \times 30,000)}{72,800 + 142,500 + 20,000 + 30,000}$$

$$= 96.5 \text{ in. above top of rail (ATR)}$$

3.6 Opening and Closing of Doors

Mechanical trucks (fork lifts, etc.) must not be used to open or close freight car doors. If doors cannot be opened, contact the serving railroad for assistance. Ensure that doors can be fully closed and properly locked. For more information regarding the proper use and care of boxcar doors, see http://www.aar.com/standards/damage-training.php.